

SC- 366-250



**Power
Generation**

Cummins India Ltd.

Power Generation Business Unit

Engine Test Certificate

QSB7-G5

This is to certify that Engine Model

84496030

Serial Numberwas tested using

Diesel as per ISO - 3046 Standard.

Corrected Full Load B.H.P to N.T.P conditions as per the

test standard is..... B.H.P. @R.P.M.
285 1500
324BHP @ 1800RPM

Date :

29-Jun-18

Product Conformance

A048U279

Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N18H322702**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
 Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
 Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: UCI274H1	AVR: SX460	AVR NO: 1107671878	ENCLOSURE: IP23	
RATED KVA: 200	RATED VOLTS: 400	RATED AMPS: 288.7	FREQUENCY: 50	PHASE: 3
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Base Continuous	WIRE: 4-Wire
EXC. VOLTS: 58	EXC.AMPS: 2.3	AMB TEMP: 40		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.03371	2+	OK	H
ROTOR	1.9663	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	12	.5	0.00
50	400	288.7	200	.8	160	58	2.3	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.UC27.4.H.1.311.,3,,11.5.,SX460.Black - RAL9011
 Remarks:

Date: 11-AUG-18	Job No: 3540507-001	AUTHORISED BY Shrikant B Labade Quality Assurance
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Certificate Of Conformity

Quality System Approval
 ISO 9001/2008

Frame Size : **UCI274H1**Serial Number : **N18H322702**

Manufactured Date : 11-AUG-18

KVA: **200**Authorized By Quality Assurance: **Shrikant B Labade**

EU DECLARATION OF CONFORMITY



**Generator
Technologies**

This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

2014/35/EU

Low Voltage Directive

2014/30/EU

The Electromagnetic Compatibility (EMC) Directive

and that the standards and/or technical specifications referenced below have been applied:

EN 61000-6-2:2005

Electromagnetic compatibility (EMC). Generic standards - Part 6-2:
Immunity for industrial environments

EN 61000-6-4:2007+A1:2011

Electromagnetic compatibility (EMC). Generic standards - Part 6-4:
Emission standard for industrial environments

EN ISO 12100:2010

Safety of machinery - General principles for design - Risk assessment
and risk reduction

EN 60034-1:2010

Rotating electrical machines - Part 1: Rating and performance

BS ISO 8528-3:2005

Reciprocating internal combustion engine driven alternating current
generating sets - Part 3: Alternating current generators for generating sets

BS 5000-3:2006

Rotating electrical machines of particular types or for particular
applications - Part 3: Generators to be driven by reciprocating internal
combustion engines - Requirements for resistance to vibration

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Limited, 49/51 Gresham Road, Staines, Middlesex, TW18 2BD, U.K.

Signed:

Date: 01st February 2016

Name, Title and Address:

Kevan J Simon
Global Technical and Quality Director
Cummins Generator Technologies
Fountain Court
Lynch Wood
Peterborough, UK
PE2 6FZ

Description UCI274H1

Serial Number N18H322702

450-16383-F

	GENERATING-SET LOAD TEST		Model No: PI 250 C 60 Hz	
			Type: Close Job No: 9522	
			Color: White Date: 01/11/2018	

Serial Number:				
PI Model No:	PI 250 C	Prime Power:	228 [kVA]	P.F (Load): 1 Voltage: 220

Generating Set Details:

Engine:	Cummins	Model No: QSB7 - G5	Serial No: 84496030
Alternator:	Stamford	Model No: UCI 274 H1	Serial No: N18H322702
Controller:	DSE	Model No: 6120	Serial No: 6865752
C. Breaker:	Metasol	Model No: ABN 803C	Serial No: N/A
C.Transformer:	Ziegler 800/5A	Model No: N/A	Serial No: N/A
AVR:	Stamford	Model No: SX 460	Serial No: 110767 - 1878
Cooling System: Water Cool	Enclosure Type: Canopy	Protection Class: IP23	

Load Test Ambient Conditions:

Temperature	33 [°C]	Altitude:	0 [m]	Humidity:	40 %
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Generating Set Rating:

Testing Full Load Current:	478.68 Amps	Nominal Current:	598.3 Amps
Power Correction Factor (De-rating):	0.000 %	Corrected Prime Power:	0.0 [kVA]

LOAD TEST

Load	Run Time	Freq	Speed	VOLTAGE L-L			CURRENT			Power	Coolant Temp	Oil Pressure	Charg-Alt.
				L1	L2	L3	L1	L2	L3				
%	min	Hz	RPM	V	V	V	A	A	A	kW	C	bar	VDC
0 %	5	60	1800	220	220	220	-	-	-	-	62	4.64	13.9
25 %	5	60	1800	219	220	220	118	117	117	44.0	70	4.04	13.9
50 %	5	60	1800	219	220	220	240	240	239	91.0	79	3.80	13.9
75 %	5	60	1800	220	220	221	362	362	362	138.0	84	3.64	13.9
100 %	10	60	1800	220	220	221	474	475	474	181.0	86	3.52	13.9
110 %	10	60	1800	220	220	221	523	526	522	200.0	88	3.44	13.9

LOAD ACCEPTANCE TEST

Suddenly Block Load %	Voltage [V]	Frequency [Hz]	Current [A]	Response Time [sec]
50%	220	60.0	240	0
0%	225 - 220	61.0 - 60.0	0	2
50%	212 - 220	57.8 - 60.0	242	2

PROTECTION SYSTEM TEST

Over Frequency Warning / SD	Under Frequency Warning / SD	Over Voltage Warning / SD	Under Voltage Warning / SD	Oil Pressure SD	Coolant Level SD	Coolant Temp Warning / SD
65 / 67	55 / 53	240 / 260	190 / 180	1.0	ECM OK	103 / 108 c



TESTED

Testing Department

ACCESSORIES DETAILS	
Model:	Serial No.
Model:	Serial No.

PI Document no.	PI-TST-LT-01	Rev.	01
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Head office :
500 Jackson Street
Columbus, IN
47201, U.S.A.

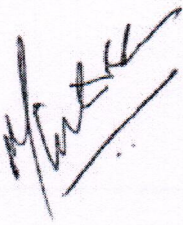
**2006/42/EC MACHINERY DIRECTIVE
DECLARATION OF INCORPORATION OF PARTLY
COMPLETED MACHINERY
Original Version**

Function:	Engine/Radiator Assembly	Model name:	QSD2-G5
Serial Number:	84496030	Date:	JUNE-2018

The partly completed machinery supplied with this declaration:

- Is designed and constructed solely as a non-functional component to be incorporated into a machine requiring completion.
- Must not be put into service within the European Community ("EC") until the final machinery into which it is to be incorporated has been declared in conformity with the Machinery Directive and all other applicable EC Directives.
- Is designed and constructed to comply with the essential health and safety requirements of the Machinery Directive 2006/42/EC listed overleaf.

The relevant technical documentation is compiled in accordance with the provisions of part B of Annex VII of the Machinery Directive. All relevant information about the partly completed machinery will be provided, in writing, on a reasoned request by the appropriate national authority to its authorised representative. The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Power Generation Limited, 49/51 Gresham Road, Staines, Middlesex, TW18 2BD, U.K.

Signed :- 	Name, Title and Address:- Rahul Thete, Plant Manager - Director, Cummins India Limited - Power System Plot No. B-2, SEZ Industrial Area, Village: Nandal - Surwadi, Taluka-Phaltan, Satara-415522, Maharashtra, India
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